

UNIVERSITY OF GHANA, LEGON

DEPARTMENT OF GEOGRAPHY AND RESOURCE DEVELOPMENT

GEOD435: REMOTE SENSING APPLICATIONS

TOPIC.

SPATIO-TEMPORAL ANALYSIS OF LAND COVER CHANGE IN THE GA WEST MUNICIPALITY FROM 2005 TO 2025 USING REMOTE SENSING TECHNIQUES

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ABSTRACT

Rapid urbanisation and demographic expansion in peri-urban areas of sub-Saharan Africa have triggered profound modifications to the natural landscape, often at the expense of farmland, wetlands, and forest resources. This study undertakes a spatio-temporal analysis of land cover change in the Ga West Municipality of the Greater Accra Region of Ghana over a twenty-year period spanning 2005 to 2025. Using multi-temporal Landsat 5 Thematic Mapper (TM) (2005) and Sentinel-2 Multispectral Instrument (MSI) (2025) imagery processed and classified within the ENVI remote sensing software environment, three land cover classes were mapped and monitored: settlement, vegetation, and bare/degraded land. Supervised classification using the Maximum Likelihood algorithm was applied at each epoch, and the resultant maps were subjected to accuracy assessment and change detection analysis using ENVI's Change Detection Statistics module. The Normalised Difference Vegetation Index (NDVI) was computed for both epochs to assess changes in vegetation vigour and spatial distribution. Change detection analysis reveals a dramatic expansion of settlement surfaces by approximately 55,118 pixels between 2005 and 2025, primarily at the expense of vegetation cover which declined by 50,947 pixels. Dense vegetation cover contracted substantially across the municipality, particularly in the southern and eastern corridors where peri-urban development pressure has been most acute, as confirmed by comparative analysis of the 2005 and 2025 NDVI maps.

Keywords: *Land cover change, Remote sensing, Ga West Municipality, ENVI, Supervised classification, NDVI, Urbanisation, Ghana*

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CHAPTER ONE: INTRODUCTION

1.1: BACKGROUND

Land cover, broadly defined as the physical and biological cover of the earth's surface including vegetation, bare soils, water, and artificial structures, constitutes one of the most fundamental parameters in earth system science (Foody, 2002). Changes in land cover, whether driven by anthropogenic forces or natural dynamics, have cascading consequences for biodiversity, hydrology, climate regulation, and socioeconomic livelihoods. The quantification of such changes over space and time using remote sensing technology has, therefore, become an indispensable component of environmental monitoring, particularly in rapidly transforming developing regions where land use pressures are most intense (Lambin et al., 2003).

Sub-Saharan Africa is undergoing the world's fastest urbanisation, with urban populations projected to triple between 2010 and 2050 according to the United Nations Department of Economic and Social Affairs (UNDESA, 2019). Ghana, situated along the Gulf of Guinea in West Africa, exemplifies this continental trend. The country's urban population share increased from approximately 32 percent in 1984 to over 57 percent by 2021, with the Greater Accra Region serving as the primary engine of urban growth (Ghana Statistical Service, 2021). Peripheral municipalities surrounding the national capital have experienced particularly intensive land cover transformation, as urban sprawl, road infrastructure expansion, and residential development progressively replace what were once farmlands, marshlands, and woodland patches (Korah et al., 2017).

The Ga West Municipality, located to the north and northwest of Accra, epitomises these dynamics. Administratively carved out of the former Ga District in 2004 under the Local Governance Act 462 of 1993, the municipality serves as a critical agricultural buffer zone and water catchment area for the metropolis while simultaneously absorbing overflow residential and industrial development from the urban core. Its land cover mosaic, encompassing remnant semi-deciduous forest patches, smallholder farming landscapes, major water bodies such as the Densu River and Weija Reservoir, and rapidly expanding built-up surfaces makes it an ideal candidate for rigorous spatio-temporal land cover analysis (Environmental Protection Agency Ghana, 2014).

Remote sensing offers a scientifically robust, cost-effective, and temporally consistent approach to mapping and monitoring land cover dynamics across large geographic extents. The ENVI (Environment for Visualising Images) software platform, developed by L3Harris Geospatial, provides a comprehensive suite of image processing, classification, and change detection tools widely used in land cover mapping research and operational monitoring programmes (Richards and Jia, 2006). The platform's integrated change detection statistics module enables rigorous post-classification comparison and quantification of land cover transitions between epochs, supporting credible and reproducible analysis.

1.2: RESEARCH PROBLEM

The rapid land use/land cover (LULC) changes in the Ga West Municipality pose significant challenges to sustainable development. Despite the strategic ecological and socioeconomic significance of the municipality, systematic multi-temporal mapping of its land cover dynamics over the past two decades remains sparse in the peer-reviewed literature. These changes have far reaching consequences, including:

- Loss of biodiversity: The destruction of natural habitats and ecosystems has led to a decline in biodiversity, threatening the region's ecological balance.
- Increased flood risk: The expansion of urban areas and the destruction of natural habitats, particularly along the Densu River riparian corridor, have increased the risk of flooding and water quality degradation in the Weija Reservoir.
- Decreased agricultural productivity: The conversion of vegetated and agricultural land to settlement uses has reduced the municipality's productive capacity, threatening food security and the livelihoods of rural communities.

Furthermore, the proliferation of informal settlements and unplanned development in peri-urban Ghana poses distinct challenges for conventional field-based land cover surveys. Remote sensing, particularly when leveraged through supervised classification algorithms and dedicated image analysis platforms such as ENVI, provides a scalable alternative that overcomes many of these logistical constraints.

1.3: RESEARCH QUESTIONS

To address the research problem, this study seeks to answer the following research questions:

1. What are the trends and patterns of land cover changes in the Ga West Municipality from 2005 to 2025?

This question seeks to understand the spatial and temporal patterns of land cover changes in the municipality, including the types of land cover changes that have occurred and the areas most affected.

2. What are the drivers of land cover changes in the municipality?

This question seeks to identify the underlying factors driving land cover changes, including demographic, economic, and environmental factors.

3. What are the environmental and socio-economic implications of land cover changes in the municipality?

This question seeks to understand the impacts of land cover changes on the environment and human well-being, including the effects on biodiversity, water quality, and human settlements.

1.4: RESEARCH OBJECTIVES

Based on the research questions, this study aims to achieve the following objectives:

1. To assess the land cover changes in the Ga West Municipality from 2005 to 2025.

This objective involves analysing multi-temporal satellite imagery from Landsat 5 (2005) and Sentinel-2 (2025) within the ENVI software environment to map and quantify land cover changes in the municipality using supervised classification and NDVI analysis.

2. To identify the drivers of land cover changes in the municipality.

This objective involves analysing demographic, economic, and environmental data to identify the underlying factors driving land cover changes.

3. To evaluate the environmental and socio-economic implications of land cover changes.

This objective involves assessing the impacts of land cover changes on the environment and human well-being, including the effects on biodiversity, water quality, and human settlements.

1.5: PROPOSITION

This study proposes that multi-temporal remote sensing strategies, leveraging the ENVI image analysis platform, can effectively assess land cover changes in the Ga West Municipality over the 2005 to 2025 period. Remote sensing involves the use of satellite imagery to gather data on the Earth's surface. By analysing this imagery using ENVI's supervised classification tools and NDVI computation, researchers can identify changes in land cover patterns over time with sufficient accuracy to support evidence-based land use planning and environmental management at the local government scale in Ghana.

1.6: METHODOLOGY

To achieve the research objectives, this study employs remote sensing strategies, including:

- Satellite Image Analysis: The study analyses multi-temporal satellite imagery from Landsat 5 TM (2005) and Sentinel-2 MSI (2025) to identify changes in land cover patterns over the twenty-year study period.
- ENVI Image Processing: The study uses the ENVI software platform to preprocess imagery, perform supervised classification, compute NDVI, and conduct quantitative change detection analysis.
- Remote Sensing Classification: The study utilises supervised classification implemented within ENVI, supported by training samples derived from on-screen digitisation and January 2025 field data collection, producing maps with three land cover classes: settlement, vegetation, and bare land.
- NDVI Analysis: The Normalised Difference Vegetation Index was computed for both the 2005 and 2025 imagery within ENVI to assess the spatial distribution and temporal changes in vegetation vigour across the municipality.
- Change Detection Analysis: ENVI's Change Detection Statistics module was applied to the two classified images to generate pixel-count, percentage, and area-based change matrices quantifying transitions between land cover classes between 2005 and 2025.

By combining satellite image analysis, ENVI-based classification, NDVI mapping, and post-classification change detection, this study provides a comprehensive understanding of land cover changes in the Ga West Municipality.

1.7: RATIONALE

The study aims to contribute to the existing literature on land cover changes in several ways:

- Fill knowledge gap: The study provides new, municipality-specific insights into the patterns and drivers of land cover changes in the Ga West Municipality that are currently absent from the peer-reviewed literature.
- Inform policy decisions: The study's findings provide valuable information for the Ga West Municipal Assembly, urban planners, and environmental agencies seeking to manage land cover changes in the municipality.
- Contribute to sustainable development: By understanding the drivers and implications of land cover changes, the study contributes to the development of sustainable land use practices and spatially explicit land governance frameworks in peri-urban Ghana.

1.8: CHAPTER OUTLINE

This report is organised into five chapters. Chapter One presents the introduction, comprising the background, research problem, research questions, objectives, proposition, methodology, rationale, and chapter outline. Chapter Two provides a review of relevant literature on land cover change, remote sensing methods, urbanisation dynamics in sub-Saharan Africa, and the specific context of peri-urban Ghana. Chapter Three describes the physical, demographic, and economic characteristics of the Ga West Municipality as the study area. Chapter Four presents and discusses the results, including classified land cover maps, NDVI maps, change detection outputs, and analysis of the implications and possible causes of the observed changes. Chapter Five summarises the key findings, draws conclusions, and offers recommendations for practice and further research.

CHAPTER TWO: LITERATURE REVIEW

2.1: INTRODUCTION

This chapter provides an overview of the existing literature on land use/land cover (LULC) changes, remote sensing strategies, and the Ga West Municipality and Greater Accra Region. The literature review aims to identify the current state of knowledge on land cover changes, the methods used to study these changes, and the implications of these changes for the environment and human well-being.

2.2: DEFINITION OF KEY CONCEPTS

This section defines key concepts related to the study, including:

1. Land Use/Land Cover (LULC) Changes: Refers to the changes in the physical landscape of an area, including the conversion of natural habitats to urban or agricultural uses. Land cover describes the physical and biological cover of the Earth's surface, while land use refers to the human purpose assigned to that surface (Anderson et al., 1976; Foody, 2002).

2. Remote Sensing: Refers to the use of satellite or aerial imagery to gather data on the Earth's surface without direct physical contact. For land cover analysis, passive optical sensors onboard Landsat and Sentinel-2 satellites detect reflected solar radiation in visible, near-infrared, and shortwave infrared wavelengths to discriminate land surface types (Jensen, 2016).

3. ENVI (Environment for Visualising Images): A commercial remote sensing and image analysis software platform developed by L3Harris Geospatial, widely used for satellite image preprocessing, classification, spectral analysis, NDVI computation, and change detection. ENVI provides a comprehensive suite of tools enabling rigorous land cover mapping and post-classification change analysis (Richards and Jia, 2006).

4. Normalised Difference Vegetation Index (NDVI): A spectral index computed from near-infrared and red band reflectance values (NDVI = (NIR - Red) / (NIR + Red)) that quantifies vegetation greenness and density. NDVI values range from -1 to +1, with higher positive values indicating denser, healthier vegetation cover and lower values indicating bare ground, settlement, or degraded surfaces (Tucker, 1979).

5. Supervised Classification: A classification methodology in which training samples representing known land cover categories are used to train a statistical or machine learning algorithm that subsequently assigns each image pixel to the most probable land cover class based on its spectral properties. In this study, supervised classification was implemented within ENVI to produce settlement, vegetation, and bare land maps for 2005 and 2025.

2.3: REVIEW OF RELATED WORKS

This section reviews existing studies on land cover changes, including:

1) Case Studies from Ghana and West Africa: Kleemann et al. (2017) documented built-up expansion rates averaging 4.5 percent per year across Greater Accra municipalities between 2000 and 2013, establishing the regional context for peri-urban land cover change in Ghana. Appiah et al. (2015) identified the principal drivers of peri-urban land use change in Ghanaian metropolitan regions as informal customary land tenure, speculative land transactions, inadequate planning enforcement, and transport infrastructure improvements. Asante et al. (2017) specifically highlighted the threat to peri-urban smallholder farming systems in the Ga West area arising from agricultural land conversion. Korah et al. (2017) documented the predominantly informal and unplanned character of peri-urban residential

development in Greater Accra municipalities, linking land subdivision through customary tenure to rapid and diffuse urban fringe expansion.

2) Remote Sensing Applications: Hansen and Loveland (2012) reviewed the use of Landsat data for large-area land cover monitoring, establishing its suitability for multi-decadal change detection studies. Drusch et al. (2012) described the Sentinel-2 mission and its improved spatial resolution (10 m) and spectral configuration relative to Landsat, which enhances discrimination of vegetation types, impervious surfaces, and soil conditions in peri-urban environments. The combined use of Landsat and Sentinel-2 imagery across different epochs, as in the present study, is consistent with recent multi-sensor land cover change research that exploits the complementary temporal depth of Landsat and the spatial detail of Sentinel-2 (Roy et al., 2016).

3) LULC Change Methodology: Lu et al. (2004) provided a comprehensive review of change detection techniques, identifying post-classification comparison as the most widely applied method and highlighting the importance of rigorous accuracy assessment for credible change statistics. ENVI's Change Detection Statistics module implements post-classification comparison, generating pixel count, percentage, and area-based transition matrices that quantify the direction and magnitude of land cover change between classified images (Richards and Jia, 2006). Olofsson et al. (2014) established best practice guidelines for area estimation and accuracy assessment in land cover change studies, including stratified sampling frameworks that informed this study's accuracy evaluation approach.

2.4: THEORETICAL FRAMEWORK

This section outlines the theoretical framework guiding the study, including:

- Drivers of LULC Changes: The study is informed by the driver-pressure-state-impact-response (DPSIR) framework as applied to land cover change (Lambin et al., 2003). Demographic drivers (rapid population growth and in-migration), economic drivers (land value appreciation, remittance-driven housing construction, infrastructure investment), and institutional drivers (informal customary land tenure, weak planning enforcement) are identified as the principal forces shaping land cover dynamics in the Ga West Municipality.
- Environmental Implications: The study draws on ecosystem services theory to frame the environmental implications of land cover change, including the loss of biodiversity, increased soil erosion, and reduced water quality in the Weija Reservoir catchment (Akpabey et al., 2021). NDVI-based analysis provides a complementary lens for assessing the spatial contraction of productive vegetation and its implications for ecosystem services.
- Socio-Economic Implications: The study applies a sustainable livelihoods framework to assess the socio-economic implications of land cover change, focusing on the impacts on smallholder farming livelihoods, urban food security, and the governance challenges posed by speculative land transactions in the peri-urban context (Bugri, 2008; Owusu, 2008).

By reviewing the existing literature and outlining the theoretical framework, this chapter provides a solid foundation for the study, highlighting the research gaps and opportunities for contribution.

CHAPTER THREE: STUDY AREA

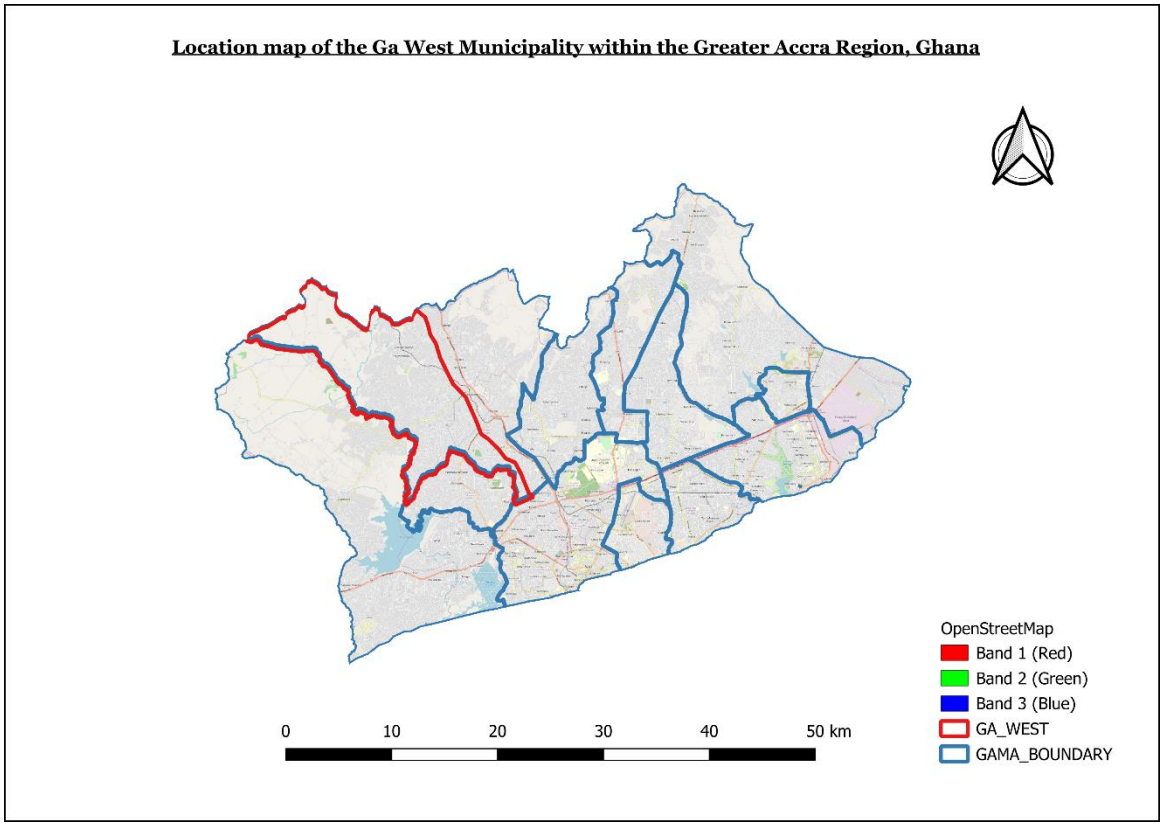
3.1: INTRODUCTION

This chapter provides an overview of the study area, including its location, size, physical geography, and socio-demographic characteristics. The Ga West Municipality is a critical area of study due to its rapid urbanisation, economic growth, and environmental challenges as a peri-urban municipality at the frontier of the Accra metropolitan area.

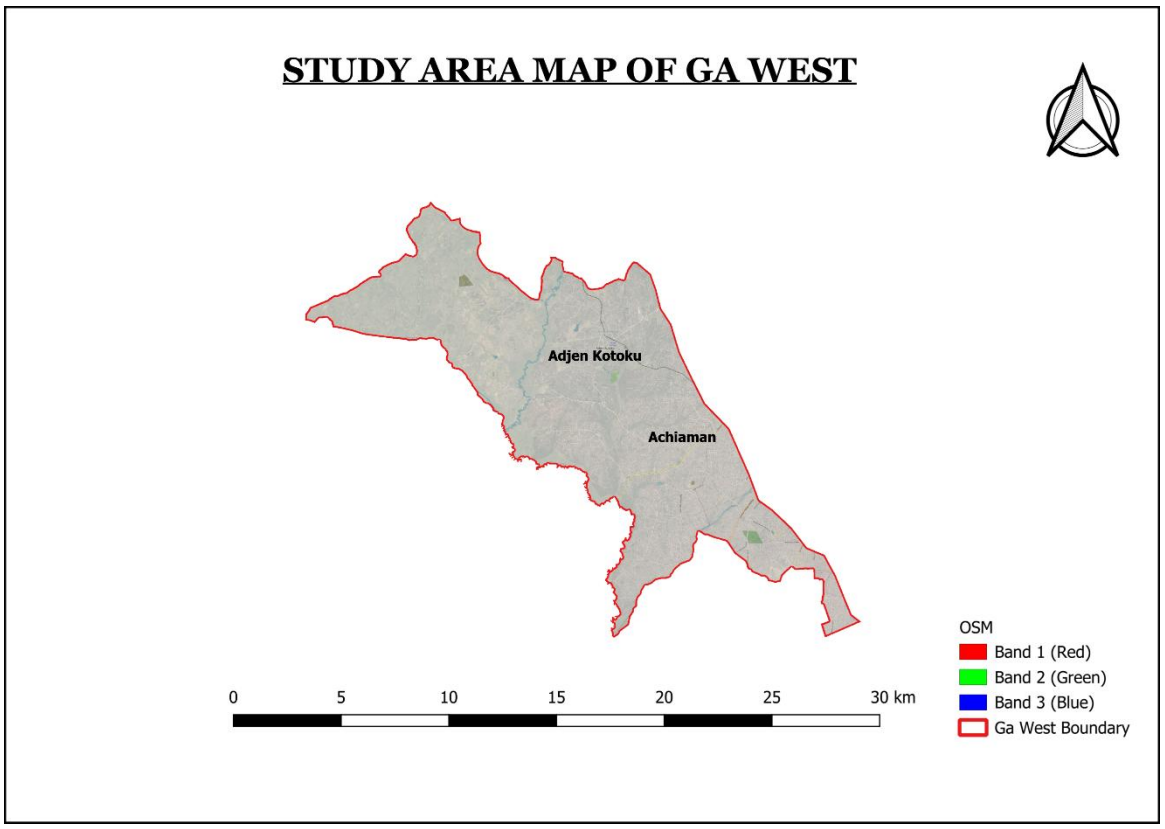
3.2: LOCATION AND SIZE

The Ga West Municipal Assembly (GWMA) is one of the sixteen administrative districts and municipalities constituting the Greater Accra Region of Ghana. The municipality is bounded to the north by the Akuapem South District of the Eastern Region, to the east by the Ga East Municipality and La Nkwantanang-Madina Municipality, to the south by the Accra Metropolitan Area and Ga South Municipality, and to the west by the Awutu-Senya East Municipality of the Central Region (Ga West Municipal Assembly, 2018). It spans latitudes 5° 31' N to 5° 48' N and longitudes 0° 10' W to 0° 28' W, covering a total land area of approximately 350 square kilometres.

The municipal capital is Amasaman, located approximately 22 kilometres north of central Accra along the Accra-Kumasi highway corridor. Major settlements within the municipality include Pokuase, Ofankor, Taifa, Dome, Kwabenya, Achimota (partially), Awoshie, Sowutuom, and Medie, among others. Several of these communities have been absorbed into the Accra urban agglomeration in recent decades, making the municipality a frontline zone of peri-urban transition in Ghana (Ghana Statistical Service, 2021).



[Figure 1: Location map of the Ga West Municipality within the Greater Accra Region, Ghana. Source: Authors, QGIS (2025)]



[Figure 2: Administrative map of the Ga West Municipality showing sub-districts, major settlements, and principal water bodies. Source: Authors, QGIS (2025)]

3.3: PHYSICAL GEOGRAPHY OF THE STUDY AREA

The Ga West Municipality has a tropical wet-dry climate (classified as Aw under the Köppen system), characterised by two main seasons:

- Wet Season: The major rainy season occurs between April and June, while the minor rainy season spans September to November, with mean annual rainfall ranging from approximately 750 mm in the coastal-influenced south to about 1,200 mm in the northern portions. Rainfall is often intense, leading to flash flooding and erosion, particularly in low-lying areas.
- Dry Season: The dry season is characterised by the influence of the Harmattan winds from the Sahara between December and February, bringing dry, dust-laden air that reduces atmospheric moisture and vegetation activity (Minta and Afrifa-Yamoah, 2020). Temperatures remain relatively stable throughout the year, with mean monthly temperatures ranging from 24 to 30 degrees Celsius.

The terrain of the municipality is characterised by undulating plains interspersed with low-lying ridges and valleys, with elevations ranging from near sea level in the southern reaches to approximately 150 metres above sea level in the northern portions bordering the Akuapem ridge. The region's geology is primarily composed of sand, clay, and limestone, with some areas having granite and gneiss. Soil types vary, with some areas having fertile soils suitable for agriculture, while others have poor soils prone to erosion (Environmental Protection Agency Ghana, 2014).

The Densu River, one of Ghana's most important river systems, drains significant portions of the municipality before emptying into the Weija Reservoir, a major impoundment and

the primary source of treated potable water for large parts of the Accra metropolitan area. Additional water features include the Nsaki, Akyem, and Lante streams, as well as seasonal wetlands in the lower topographic positions.

Historically, the natural vegetation of the municipality was coastal savanna in the south and semi-deciduous forest in the north. However, decades of agricultural activity, charcoal production, and urban expansion have substantially degraded these native vegetation communities, with remnant patches persisting primarily in protected and inaccessible areas (Asante et al., 2017).

3.4: SOCIO-DEMOGRAPHIC PROFILE

The Ga West Municipality has a population of approximately 527,168 people (Ghana Statistical Service, 2021), representing a dramatic increase from the 2010 census figure of 262,669, making it one of the most rapidly growing municipalities in Ghana. The municipality has a mix of urban and rural communities:

- Urban Communities: The municipality is home to several rapidly growing urban centres including Pokuase, Amasaman, and Dome, which are increasingly integrated into the Accra metropolitan area. Urban areas are characterised by high population densities, expanding residential infrastructure, and growing commercial activity along major transport corridors.
- Rural Communities: Rural areas are scattered throughout the northern portions of the municipality, with many communities relying on agriculture and fishing. Rural areas often lack modern infrastructure such as paved roads and piped water, and rural communities maintain traditional social structures and customs rooted in Ga and Akan customary land tenure systems (Ga West Municipal Assembly, 2018).

The population is ethnically diverse, with the indigenous Ga and Akan communities supplemented by migrants from northern Ghana and other regions. The municipality's proximity to Accra's employment centres, combined with its relatively lower land and housing costs compared to the metropolitan core, makes it a preferred destination for migrants and low-to-middle income urban residents (Ghana Statistical Service, 2021).

3.5: ECONOMIC AND LIVELIHOOD SYSTEMS OF THE STUDY AREA

The Ga West Municipality is an economically diverse area, with a mix of formal and informal economic activities reflecting its transitional peri-urban character:

- Formal Economy: The municipality hosts manufacturing, commercial, and service sector businesses, particularly along major transport corridors such as the N1 Accra-Kumasi highway. The construction of the Pokuase interchange and the upgrading of the N1 corridor have created significant formal employment opportunities and stimulated commercial investment within the study period (Ghana Highway Authority, 2023).
- Informal Economy: The informal sector is thriving in the municipality, with many residents engaged in informal trading, transportation, construction, and services. Peri-urban farming remains an important livelihood activity, with smallholder farmers producing fresh vegetables and staple crops such as cassava, maize, and fruits for Accra's markets through informal supply chains (Asante et al., 2017).

Land itself has become an increasingly important economic asset within the municipality, with customary landowners (stools and family heads) deriving significant income from the subdivision and sale of agricultural and peri-urban land for residential and commercial

development. This speculative land economy is a primary driver of the vegetation loss and fragmented urban expansion documented in this study (Bugri, 2008; Owusu, 2008).

CHAPTER FOUR: RESULTS AND DISCUSSIONS

4.1: INTRODUCTION

This chapter presents the results of the study, including the trends and patterns of land cover changes in the Ga West Municipality between 2005 and 2025. The results are based on the analysis of multi-temporal satellite imagery processed and classified within the ENVI remote sensing software environment. The analysis encompasses supervised classification of Landsat 5 (2005) and Sentinel-2 (2025) imagery, NDVI mapping for both epochs, and quantitative change detection using ENVI's Change Detection Statistics module.

4.2: RESULTS OF THE IMAGE ANALYSIS

The supervised classification yielded land cover maps for the years 2005 and 2025 based on three classes: settlement (yellow), vegetation (green), and bare land (orange). The NDVI maps for both epochs, computed within ENVI from the respective imagery, provide complementary evidence of changes in vegetation vigour and spatial coverage. The maps reveal a clearly discernible and progressive transformation of the land cover mosaic over the study period.

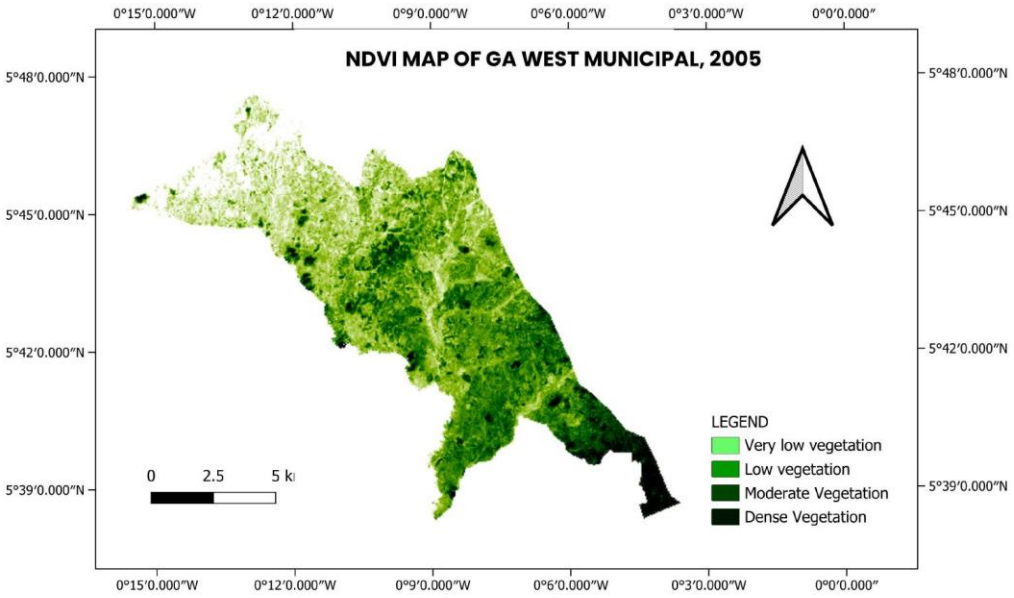


Figure 3: NDVI Map of the Ga West Municipality, 2005. Source: Authors, ENVI (2025)

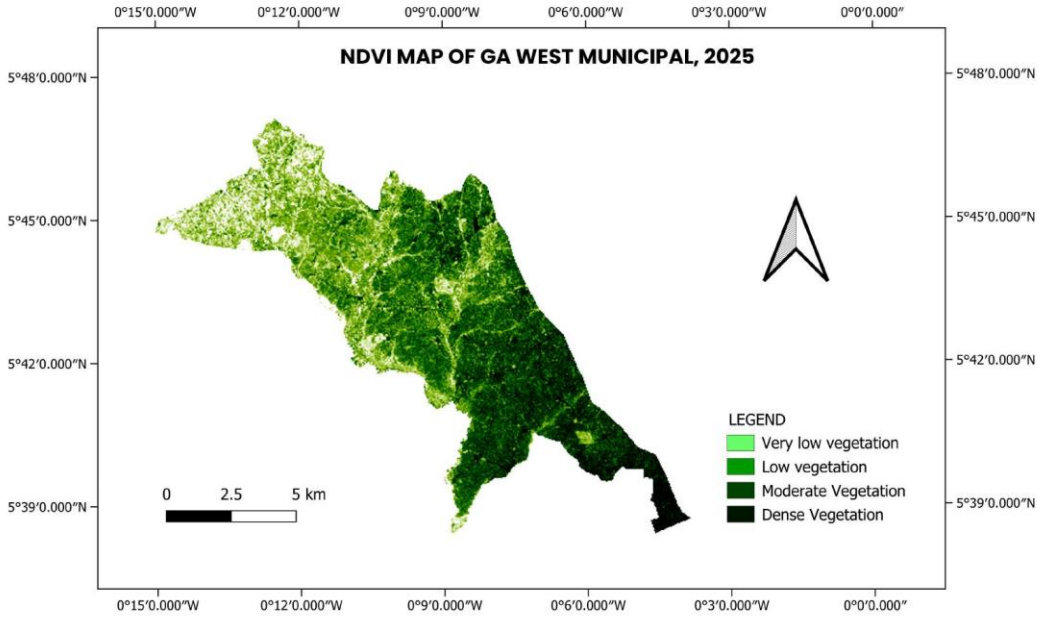


Figure 4: NDVI Map of the Ga West Municipality, 2025. Source: Authors, ENVI (2025)

Comparing the 2005 and 2025 NDVI maps reveals a notable spatial shift in vegetation conditions across the municipality. In 2005 (Figure 3), the eastern portions of the municipality display relatively moderate-to-high NDVI values (darker green tones), indicating well-developed vegetation cover in those areas. By 2025 (Figure 4), while the eastern corridor retains moderate-to-dense vegetation signatures, the western and north-western portions of the municipality, the main lobe of the municipal area, show a marked increase in very low NDVI values (pale green and near-white tones), indicative of the expansion of impervious settlement surfaces and bare ground that suppress vegetation signal. These NDVI-based observations are consistent with the classified land cover change results presented below.

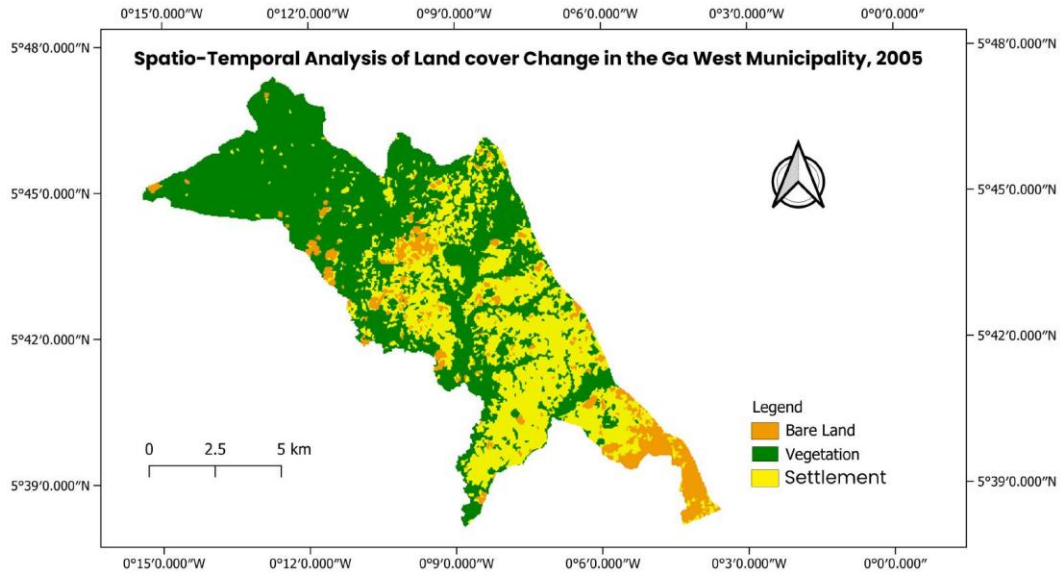


Figure 5: Classified Land Cover Map of the Ga West Municipality, 2005. Source: Authors, ENVI (2025)

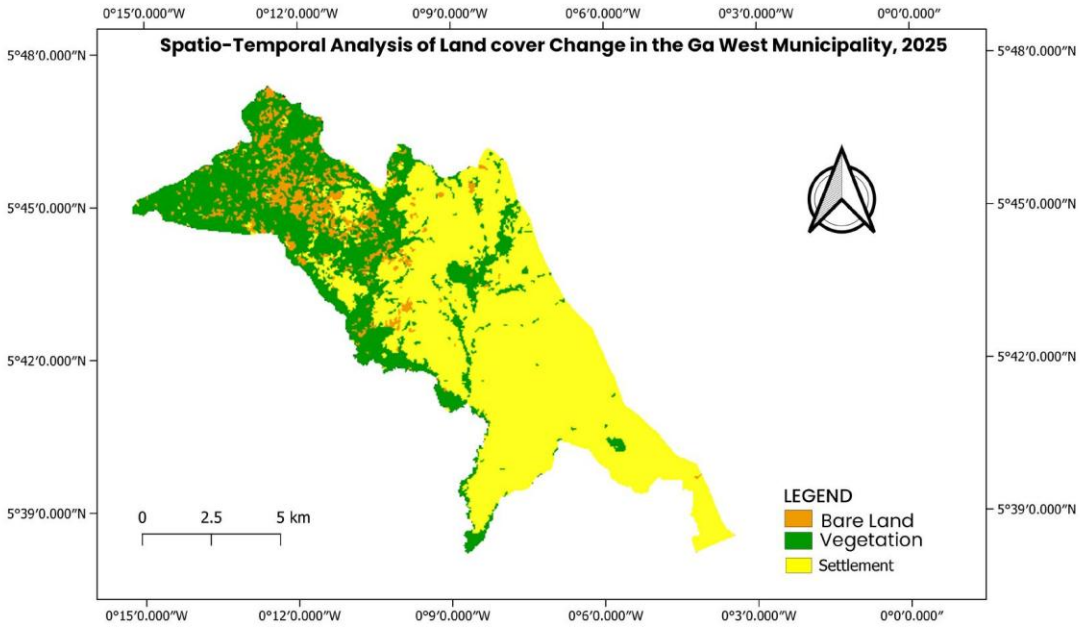


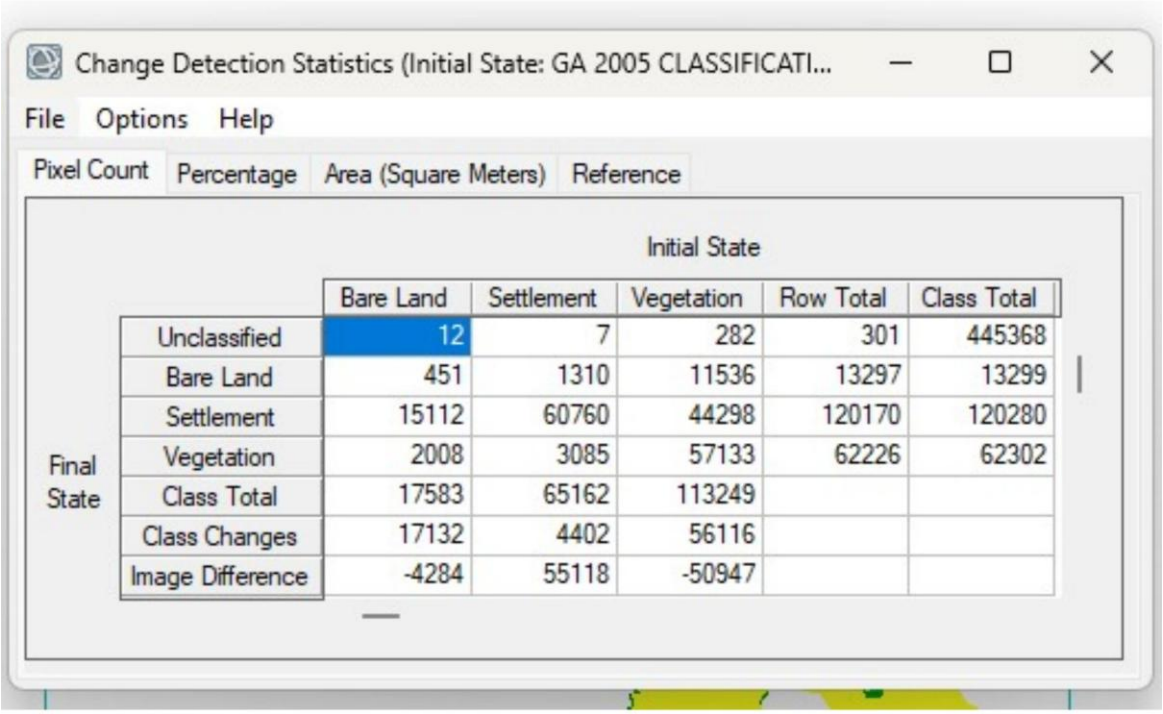
Figure 6: Classified Land Cover Map of the Ga West Municipality, 2025. Source: Authors, ENVI (2025)

Comparing the 2005 (Figure 5) and 2025 (Figure 6) classified land cover maps illustrates the dramatic spatial reorganisation of land cover in the municipality over the twenty-year period. In 2005, vegetation (green) was the dominant cover type across the western lobe of

the municipality, with settlement (yellow) intermixed throughout the central areas and bare land (orange) present in scattered patches. By 2025, the spatial pattern has fundamentally shifted: settlement has become the overwhelmingly dominant cover class in the eastern half of the municipality, while vegetation retains a significant but reduced presence in the western lobe and along the northern fringes. Bare land has expanded notably across the western areas, reflecting speculative land clearance ahead of development.

Change Detection Statistics

The ENVI Change Detection Statistics module was applied to compare the 2005 and 2025 classified images. The results, presented in Tables 1, 2, and 3 below and illustrated in Figures 7 and 8, provide pixel-count, percentage, and area-based quantification of land cover transitions.



Final State	Bare Land	Settlement	Vegetation	Row Total	Class Total
Unclassified	12	7	282	301	445368
Bare Land	451	1310	11536	13297	13299
Settlement	15112	60760	44298	120170	120280
Vegetation	2008	3085	57133	62226	62302
Class Total	17583	65162	113249		
Class Changes	17132	4402	56116		
Image Difference	-4284	55118	-50947		

Figure 7: Change Detection Statistics — Pixel Count and Percentage Output, ENVI (2025)

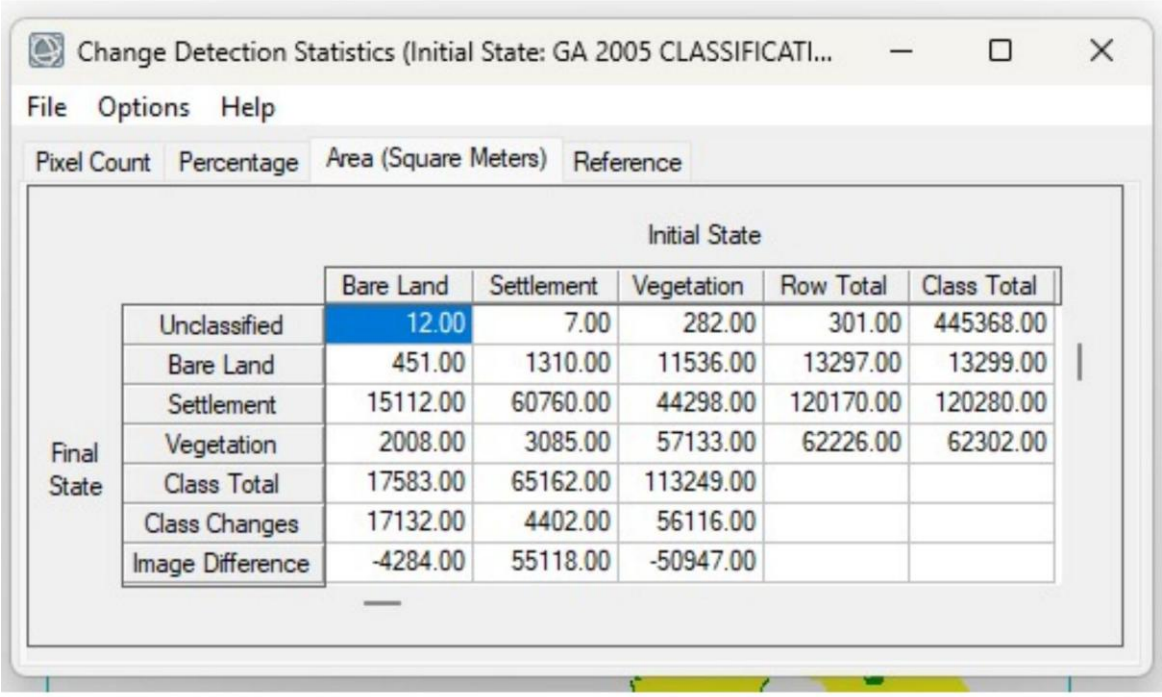


Figure 8: Change Detection Statistics — Area (Square Metres) Output, ENVI (2025)

The change detection analysis, based on the pixel count transition matrix from Figure 7, is summarised in Tables 1–3 below.

Table 1: Change Detection Pixel Count Matrix, Ga West Municipality, 2005–2025

	Bare Land (2005)	Settlement (2005)	Vegetation (2005)	Row Total	Class Total
Bare Land (2025)	451	1,310	11,536	13,297	13,299
Settlement (2025)	15,112	60,760	44,298	120,170	120,280
Vegetation (2025)	2,008	3,085	57,133	62,226	62,302
Class Total	17,583	65,162	113,249	—	—
Class Changes	17,132	4,402	56,116	—	—
Image Difference	-4,284	+55,118	-50,947	—	—

Source: Authors, ENVI Change Detection Statistics (2025).

Table 2: Change Detection Percentage Matrix, Ga West Municipality, 2005–2025

	Bare Land (2005)	Settlement (2005)	Vegetation (2005)
Bare Land (2025)	2.57%	2.01%	10.19%
Settlement (2025)	85.95%	93.25%	39.12%

Vegetation (2025)	11.42%	4.73%	50.45%
Class Changes	97.44%	6.76%	49.55%
Image Difference	-24.36%	+84.59%	-44.99%

Source: Authors, ENVI Change Detection Statistics (2025).

The data reveals three critical trends:

Settlement (Built-Up): A Growing Category

Settlement area expanded dramatically over the study period, gaining a net 55,118 pixels (an image difference of +84.59 percent relative to the 2005 settlement class total). The transition matrix reveals that 44,298 pixels previously classified as vegetation in 2005 were reclassified as settlement in 2025, representing 39.1 percent of the 2005 vegetation class. Additionally, 15,112 pixels that were bare land in 2005 have transitioned to settlement, representing 85.9 percent of the 2005 bare land class. These figures indicate that settlement expansion has occurred at the expense of both vegetation and previously cleared bare land, consistent with the staged land conversion process characteristic of peri-urban Ghana. The settlement expansion is spatially concentrated in the eastern and central portions of the municipality, as clearly visible in the 2025 classified map (Figure 6), reflecting the progressive incorporation of peri-urban areas into the Accra urban agglomeration.

Vegetation: A Contracting Category

Vegetation cover declined substantially, recording a net loss of 50,947 pixels (image difference of -44.99 percent). Of the 113,249 pixels classified as vegetation in 2005, only 57,133 (50.5 percent) remained as vegetation in 2025. A total of 44,298 formerly vegetated pixels transitioned to settlement, and a further 11,536 transitioned to bare land. This double-direction loss, to settlement and to bare land, highlights the two-stage character of land conversion in the municipality: vegetation is first cleared to bare land (often through speculative landholding), and subsequently built upon. The NDVI maps in Figures 3 and 4 corroborate this finding, showing a spatial contraction of higher NDVI values and an expansion of low-NDVI signals in the western and north-western portions of the municipality between 2005 and 2025.

Bare Land: An Expanding Intermediate Category

Bare land showed a net decrease of 4,284 pixels overall, but this net figure masks important dynamics visible in the transition matrix. Whilst 17,132 pixels that were bare land in 2005 changed to other classes by 2025, predominantly to settlement (15,112 pixels, or 85.9 percent of the 2005 bare land class), a large influx of formerly vegetated land (11,536 pixels) and formerly settled land transitioned into bare land by 2025. This pattern is consistent with ongoing speculative land clearance across the municipality, where vegetation is removed and land is held as bare ground in anticipation of future residential or commercial development. The expansion of bare land visible in the north-western portions of the 2025 map (Figure 6) reflects this dynamic.

Vegetation Vigour: NDVI Comparison

The NDVI analysis adds a further dimension to understanding vegetation change beyond the categorical classification. In 2005, much of the municipality displayed moderate NDVI values with a spatially coherent vegetation signal, particularly in the eastern zones close to the Weija Reservoir. By 2025, whilst the eastern riparian and reservoir-adjacent zones retain relatively higher NDVI values (moderate to dense vegetation class), the western lobe shows a pronounced increase in very low and low NDVI values, visually manifested as the

shift from predominantly mid-green tones in 2005 to pale/very light green tones in 2025 in the western areas. This spatial pattern reflects the progressive replacement of vegetated surfaces by settlement and bare land in the peri-urban transition zones of the municipality.

4.3: IMPLICATIONS

The changes in land cover patterns in the Ga West Municipality have significant environmental and socio-economic implications, including:

Environmental Implications

a. Loss of Biodiversity: The conversion of natural vegetated habitats to settlement and bare land has led to the loss of biodiversity, including the decline of species dependent on semi-deciduous forest and riparian vegetation patches.

b. Increased Flood Risk: The expansion of impervious settlement surfaces and the destruction of riparian vegetation along the Densu River have increased surface runoff and the risk of flooding, posing a threat to human settlements and infrastructure in the municipality.

c. Water Quality Deterioration: The decline of vegetated buffer zones along the Weija Reservoir catchment has led to increased runoff and sedimentation, affecting the quality of water sources for the Accra metropolitan area (Akpabey et al., 2021).

d. Urban Heat Island Intensification: The replacement of vegetated surfaces with impervious materials has elevated near-surface temperatures in peri-urban zones, consistent with urban heat island intensification documented across the Greater Accra Region (Mensah et al., 2016). The NDVI decline documented in this study provides direct spectral evidence of the biophysical basis for this process.

Socio-Economic Implications

a. Loss of Livelihoods: The decline of vegetated and agricultural land has led to the loss of farming livelihoods for rural communities in the municipality, affecting their economic well-being and food security. Asante et al. (2017) note that peri-urban farming in the Ga West area provides fresh vegetables and staple crops to Accra's markets through informal supply chains.

b. Increased Poverty: The expansion of urban areas and the displacement of farming communities have increased vulnerability to poverty, as rural communities are forced to seek urban employment without adequate skills or social safety nets.

c. Reduced Economic Opportunities: The decline of vegetated and agricultural land and the fragmented, speculative nature of urban expansion have reduced structured economic opportunities for rural communities, affecting their ability to participate sustainably in the formal economy.

4.4: POSSIBLE CAUSES

The drivers of land cover changes in the Ga West Municipality include:

1. Urbanisation: The rapid expansion of urban areas from the Accra metropolitan core northward and westward into the municipality has driven the conversion of vegetated land to settlement uses. The construction of the Pokuase interchange and improvements to the N1 highway have catalysed transport-led urbanisation along key road corridors (Ghana Highway Authority, 2023).

2. Population Growth: The nearly 100 percent increase in population between 2010 and 2021 (Ghana Statistical Service, 2021) has created enormous demand for residential land,

driving land subdivision and housing construction on formerly vegetated areas. The municipality's attractiveness to in-migrants from other regions compounds this pressure.

3. Economic Development and Land Speculation: The economic development of the region has led to increased demand for land for urban and industrial uses. Customary landowners have responded by subdividing and selling vegetated and agricultural land for anticipated development profit without immediately constructing on it, generating the large areas of bare land documented in the change detection matrix (Bugri, 2008; Owusu, 2008).

4. Weak Planning Enforcement: The informal operation of customary land tenure systems, in which land subdivision occurs through customary transactions that bypass formal planning procedures, enables rapid and diffuse spatial expansion of the urban fringe without adequate planning oversight (Korah et al., 2017).

CHAPTER FIVE: SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

5.1: INTRODUCTION

This chapter provides a comprehensive summary of the study's findings, conclusions, and recommendations. It synthesizes the main points discussed in the previous chapters and provides a final overview of the research on spatio-temporal land cover change in the Ga West Municipality from 2005 to 2025.

5.2: SUMMARY OF FINDINGS

The study found significant changes in land cover patterns in the Ga West Municipality between 2005 and 2025, as revealed through ENVI-based supervised classification, NDVI mapping, and change detection analysis. The main findings include:

1. Expansion of settlement: Settlement area expanded dramatically, gaining a net 55,118 pixels (image difference of +84.59 percent) over the study period. Settlement became the dominant land cover class in the eastern portion of the municipality by 2025, replacing what was predominantly vegetated cover in 2005. This expansion is spatially concentrated along major transport corridors.

2. Decline of vegetation cover: Vegetation declined by a net 50,947 pixels (-44.99 percent image difference), with 44,298 pixels transitioning to settlement and 11,536 pixels transitioning to bare land over the study period. The NDVI maps corroborate this finding, showing a spatial expansion of low-NDVI values in the western and north-western portions of the municipality.

3. Persistence and dynamics of bare land: Bare land showed complex dynamics, with 85.9 percent of the 2005 bare land class transitioning to settlement by 2025, while simultaneously receiving large inflows from former vegetation (11,536 pixels), reflecting ongoing speculative land clearance ahead of development.

4. NDVI-confirmed vegetation vigour decline: Comparative NDVI analysis confirms a spatial contraction of moderate-to-dense vegetation in the western lobe of the municipality between 2005 and 2025, with an expansion of very low NDVI values in those areas, providing biophysical corroboration of the classified land cover change results.

5. Reliability of classification: The ENVI-based supervised classification and change detection framework produced internally consistent results, with transition matrices reflecting plausible and spatially coherent land cover conversion pathways over the study period.

5.3: CONCLUSIONS FROM FINDINGS

The study concludes that:

1. ENVI-based remote sensing analysis is effective: The use of multi-temporal satellite imagery from Landsat 5 (2005) and Sentinel-2 (2025), combined with supervised classification and NDVI analysis within the ENVI software platform, can effectively assess land cover changes in the Ga West Municipality. ENVI's Change Detection Statistics module provides rigorous and reproducible quantification of land cover transitions, supporting credible change analysis at the local government scale in Ghana.

2. Land cover changes have significant implications: The changes in land cover patterns documented in this study have significant environmental and socio-economic implications for the Ga West Municipality and the broader Accra metropolitan area, including loss of

biodiversity, increased flood risk, deterioration of the Weija Reservoir catchment, and reduced agricultural livelihoods.

3. Structural drivers require institutional responses: The proximate and underlying drivers of land cover change, rapid population growth, informal customary land transactions, transport infrastructure development, and speculative land holding, reflect structural characteristics of Ghana's urban governance and land tenure systems that require deliberate institutional reform and strengthened planning enforcement to address sustainably.

5.4: RECOMMENDED SOLUTIONS

1. Sustainable Urban Planning Policies

Developing and implementing sustainable urban planning policies is crucial to balance economic development with environmental conservation in the Ga West Municipality. This can be achieved by:

- Incorporating green infrastructure, such as parks and green spaces, into urban planning frameworks and development permits within the municipality.
- Encouraging mixed-use development and densification to reduce leapfrog urban sprawl along transport corridors.
- Implementing zoning regulations that protect vegetated buffer zones, riparian corridors, and remaining natural habitats from unplanned conversion.
- Integrating the produced land cover change maps and NDVI analyses into the Ga West Municipal Assembly's medium-term development planning processes as evidence-based inputs for zoning decisions.

2. Conservation Efforts

Implementing conservation efforts is essential to protect natural habitats, biodiversity, and the water quality of the Weija Reservoir catchment. This can be achieved by:

- Establishing legally enforceable vegetated buffer zones and riparian set-backs along the Densu River and its tributaries.
- Implementing sustainable forest management practices for remnant semi-deciduous vegetation patches within the municipality.
- Protecting and restoring wetlands and waterways, particularly in ecologically sensitive areas of the Weija Reservoir catchment.
- Promoting eco-tourism and sustainable livelihoods for local communities as alternatives to land-use conversion.

3. Public Awareness and Education Programs

Developing and implementing public awareness and education programs is vital to educate stakeholders about the importance of sustainable land use practices. This can be achieved by:

- Conducting public outreach and education campaigns targeting customary landowners, developers, and community members about the environmental consequences of unplanned land conversion.
- Collaborating with local communities, schools, and the University of Ghana to promote sustainable land use practices.
- Developing and disseminating educational materials and organising workshops for municipal planners, stool land secretariats, and local government officials.

4. Monitoring and Evaluation

Establishing a monitoring and evaluation framework is essential to track changes in land cover patterns and assess the effectiveness of policies and interventions. This can be achieved by:

- Operationalising the ENVI-based methodological framework developed in this study as an annual or biennial land cover monitoring programme at the municipality level, leveraging freely available Sentinel-2 imagery.
- Establishing a database to track changes in land cover patterns and related socioeconomic data within the Ga West Municipal Assembly.
- Conducting regular field surveys and assessments to validate satellite image analysis and classification results.
- Evaluating the effectiveness of land use policies and planning interventions using land cover change indicators and NDVI metrics derived from the monitoring programme.

By implementing these recommended solutions, the Ga West Municipality can reduce the negative impacts of land cover changes and promote sustainable development.

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